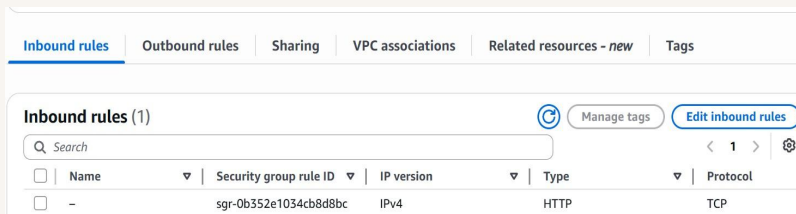

VPC Traffic Flow and Security

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Introducing Today's Project!

What is Amazon VPC?

An Amazon VPC (Virtual Private Cloud) is your own private, isolated network inside AWS where you can launch and control cloud resources like EC2 instances, databases, and load balancers. Think of it as building your own data center inside AWS — with full control over IP ranges, subnets, routing, and security.

How I used Amazon VPC in this project

In this project, I designed and deployed a fully customized Amazon Virtual Private Cloud (VPC) to learn and demonstrate how cloud networking, routing, and security layers work together inside AWS. I built a complete network environment from the ground up, including public and private subnets, routing components, and dual-layer security controls using security groups and Network ACLs. This project strengthened my understanding of how real-world cloud architectures are structured and secured.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how quickly the overall network architecture made sense once I connected all the VPC components together. Seeing traffic move from the Internet Gateway through the route tables, into the subnets, and then through both the Network ACLs and security groups gave me a much deeper understanding of how AWS enforces layered security. What seemed abstract at first became very intuitive once I built and tested it myself.

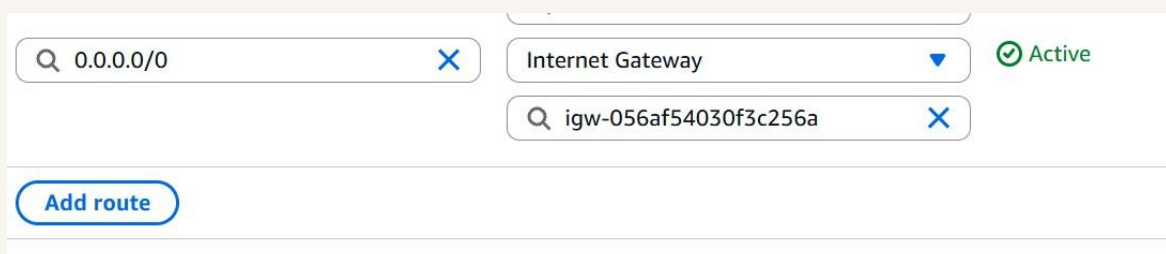
This project took me...

This project took me a few focused hours, during which I designed the VPC, created public and private subnets, configured route tables, attached an Internet Gateway, set up Network ACLs and security groups.

Route tables

A route table is how traffic moves within a VPC by specifying which destinations each subnet can reach and which targets that traffic should use. They are like GPS that directs traffic within my VPC to the correct destination. Every subnet must be associated with a route table, and the routes inside it determine whether the subnet is public, private, or isolated. Route tables are needed to make a subnet public because a subnet needs to have a route to an internet gateway in order to be considered public and is the only way to establish this connection, by adding a default route that points to the Internet Gateway, the route table enables outbound internet access for resources in a public subnet while still maintaining controlled, predictable traffic flow across the VPC.

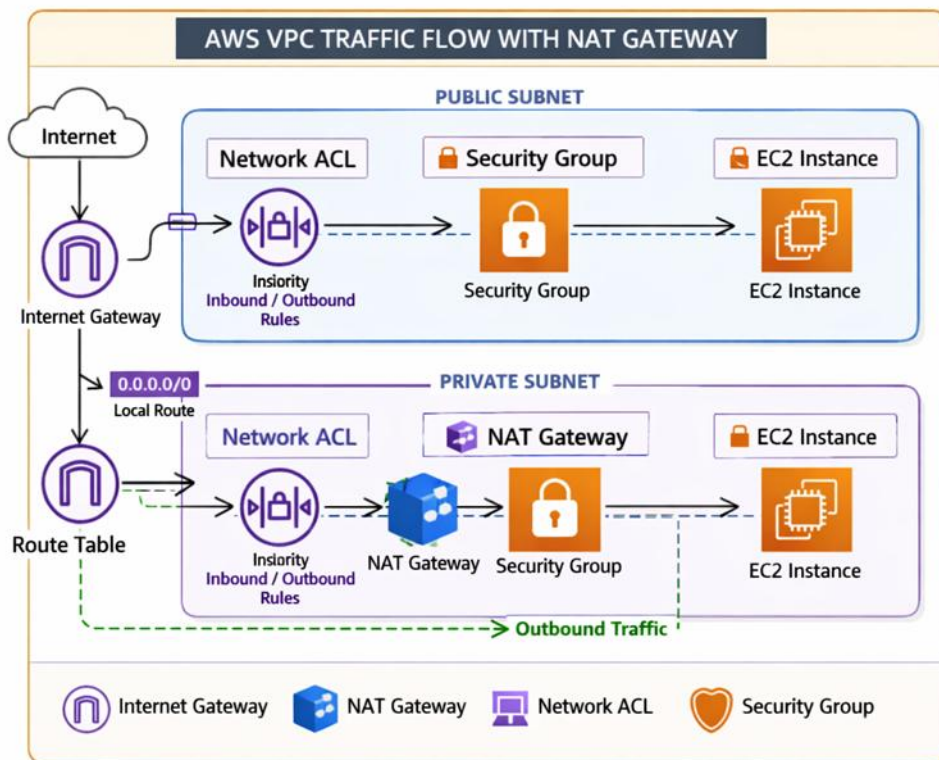
Routes tables are needed to make a subnet public because a subnet only becomes public when its route table includes a default route (0.0.0.0/0) that points to the Internet Gateway. This route gives the subnet a path to send outbound traffic to the internet and allows return traffic to flow back in. Without this entry, even if the subnet is named “public” or has auto-assign public IPs enabled, it cannot reach the internet. The route table is the component that actually provides internet connectivity, making it essential for defining a subnet as truly public.



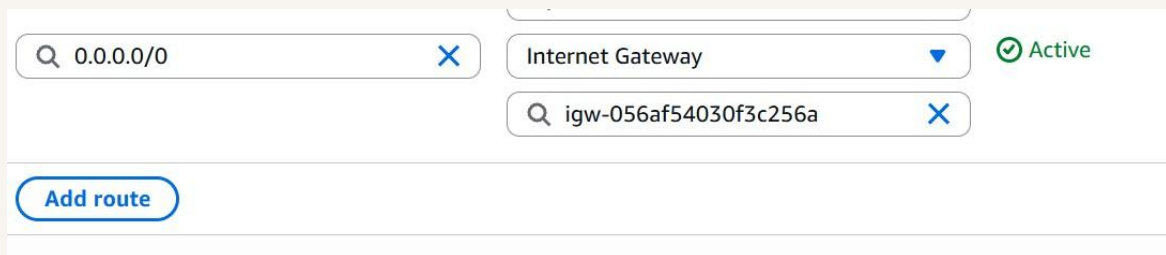
0.0.0.0/0	Internet Gateway	Active
igw-056af54030f3c256a		
Add route		

Route destination and target

Routes are defined by their destination and target, which mean two different things. The destination specifies the IP range (CIDR block) that the traffic is trying to reach, such as the VPC's internal network or all internet addresses (0.0.0.0/0). The target tells AWS where to send that traffic, such as an Internet Gateway, NAT Gateway, local VPC route, or a peering connection. Together, the destination and target determine exactly how traffic flows inside the VPC and whether it can reach external networks.



The route in my route table that directed internet-bound traffic to my Internet Gateway had a destination of 0.0.0.0/0, which represents all IPv4 addresses on the internet. The destination defines the range of IP addresses that traffic in my VPC is trying to reach, while the target specifies the path that traffic should take to get there. In this case, the target was my Internet Gateway, which provides the required path for outbound traffic to reach the internet and for return traffic to flow back into the subnet.



The screenshot shows a configuration interface for a route. It features a search bar with the text "0.0.0.0/0" and a blue "X" icon. To the right, there is a dropdown menu currently showing "Internet Gateway" with a blue downward arrow, and a green checkmark icon followed by the word "Active". Below the dropdown is another search bar containing the text "igw-056af54030f3c256a" and a blue "X" icon. At the bottom left of the configuration area is a blue button labeled "Add route".

Security groups

Security groups are responsible for checking who comes in and out. They have strict rules about what kind of traffic can enter or leave the resource based on its IP address, protocols and port numbers. Because they track the state of connections, security groups automatically allow return traffic for any request they permit, providing a controlled and predictable layer of instance-level protection.

Inbound vs Outbound rules

Inbound rules areAn inbound rule specifies what kind of traffic is allowed to reach a resource, based on the source IP address, protocol, and port number. If traffic doesn't match an inbound rule, the security group blocks it. Because security groups are stateful, once inbound traffic is allowed, the return traffic is automatically permitted. I configured an inbound rule that allows HTTP traffic on port 80 so that my instance can receive web requests from approved sources. This rule specifies that traffic using the HTTP protocol is permitted to enter the resource, and because security groups are stateful, the return traffic for these requests is automatically allowed.

An outbound rule defines what type of traffic a resource is allowed to send out, based on the destination IP range, protocol, and port number. If traffic doesn't match an outbound rule, the security group blocks it. By default, my security group's outbound rule allows all outbound traffic, meaning the resource is permitted to initiate connections to any IP address using any protocol or port. Because security groups are stateful, this default rule ensures that the instance can make outbound requests—such as software updates or API calls—and automatically receive the return traffic without needing additional inbound rules.

Inbound rules						
Inbound rules (1)						
Q Search Manage tags Edit inbound rules						
< 1 >						
☐	Name	Security group rule ID	IP version	Type	Protocol	
☐	-	sgr-0b352e1034cb8d8bc	IPv4	HTTP	TCP	

Network ACLs

Network ACLs are like community watchmen that secure my network at the subnet level, the difference between a security group and a network ACL is their scope.

Security groups vs. network ACLs

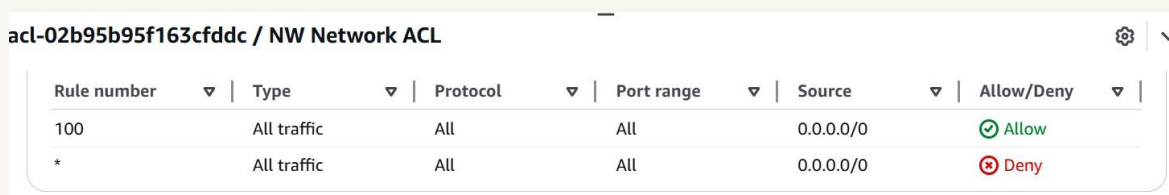
A security group secures my network resources at the resource level so every single resources in my VPC associated with a security group while a network ACL's secures my network at the subnet level every single subnet in my VPC is associated with a network ACL. Having both network ACL's and security groups is a good security practice because it creates a dual layer of security that makes sure inbound or outbound traffic go through at least 2 checks.

Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a Network ACL's inbound and outbound rules allow all traffic, meaning any protocol, any port, and any IP address is permitted to enter or leave the subnet. This open configuration ensures that new subnets function immediately, but it also means the default NACL provides no real security filtering.

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all incoming outgoing traffic. When you create a custom NACL, however, it starts with no rules at all, so all traffic is denied until you explicitly add allow rules for both inbound and outbound directions.



The screenshot shows a configuration window for a Network ACL titled 'acl-02b95b95f163cfddc / NW Network ACL'. It contains a table with two rules. The first rule, numbered 100, allows all traffic from any source. The second rule, marked with an asterisk, denies all traffic from any source.

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Tracking VPC Resources

In the us-east-2 (Ohio) Region, we created a complete networking foundation by deploying a VPC, attaching an Internet Gateway, and configuring a Security Group. This setup provides the essential components needed for launching EC2 instances and controlling how traffic flows into and out of the environment. *The VPC defines the isolated network space. *The Internet Gateway enables inbound and outbound internet connectivity for public subnets. *The Security Group provides stateful, resource-level protection for any EC2 instances launched inside the VPC. This gives you a fully functional starting point for deploying workloads in the Ohio Region. Instead of working in my usual Region, I used another AWS Region to create and deploy new resources. This helped me practice how to build infrastructure in multiple geographic locations and understand how AWS services behave across Regions. Teams use multiple Regions to improve availability, resilience, and global performance. By distributing wo

EC2 Global View is a tool where you can find all your EC2 resources across every AWS Region in one place — instances, security groups, key pairs, volumes, AMIs, and more. It gives you a single dashboard to understand what you have deployed globally, instead of checking each Region one by one. You can even narrow down your search by using filters such as Region, resource type, instance state, or tags. This makes it easy to quickly locate resources, audit your environment, or confirm where something is running.

Now that I've learned about EC2 Global View, I'd use it again anytime I need a quick, single-page view of all my EC2 resources across every Region. It's perfect for checking where instances are running, spotting resources I forgot to shut down, or filtering by Region, tags, or instance state. Without it, I'd have to manually switch Regions or run multiple CLI commands, which takes much longer.

Summary

Summary of your resources across all Regions for which your account is enabled.

Fetching resources for all opted in regions

✔ Resource update complete

Resource totals will be inaccurate until complete

Compute

4

Storage

8

Networking

368

Security

49

▼ Show all resource summary

Enabled regions

17 regions

Instances

4 in 1 regions

VPCs

19 in 17 regions

Subnets

56 in 17 regions

Security groups

28 in 17 regions

Volumes

8 in 1 regions

Auto scaling groups

0 in 0 regions

Route tables

19 in 17 regions
